

Problem-Solution Fit Canvas

Date	28 June 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1.CUSTOMER SEGMENTS[CS] The Voice-Based Concept Understanding Analyser (VBCUA) is designed for students, job aspirants, online learners, educators, and professionals who want to evaluate their conceptual understanding through spoken explanations. The primary users include engineering students preparing for exams, candidates preparing for technical interviews, teachers assessing students' conceptual knowledge, and learners seeking AI-powered feedback to improve both communication skills and subject understanding.	5. AVAILABLE SOLUTIONS Current solutions include manual teacher evaluations, mock interviews, speech recognition applications, grammar-checking tools, and online learning platforms. However, these solutions do not combine speech transcription, semantic understanding, audio feature analysis, and automated evaluation into a single intelligent platform.	8. CHANNELS OF BEHAVIOR (CH) Online: Educational platforms, YouTube, AI learning tools, coding platforms, virtual classrooms, and university learning portals. Offline: Colleges, coaching institutes, classrooms, libraries, group discussions, seminars, and technical workshops.

2. PROBLEMS/PAINS Many learners understand concepts but struggle to explain them clearly. They often receive delayed or subjective feedback and cannot accurately measure their conceptual understanding. Traditional assessments mainly evaluate written responses and rarely assess verbal explanations. Existing speech tools focus on pronunciation or grammar instead of evaluating conceptual accuracy.	6. CUSTOMER LIMITATIONS Many users have limited access to personalized mentoring, inconsistent practice opportunities, and difficulty identifying conceptual mistakes independently. Some users also have limited technical knowledge and rely mainly on desktop or mobile applications for learning.	9. PROBLEM ROOT / CAUSE (RC) The primary causes include lack of personalized feedback, difficulty in self-evaluating conceptual understanding, insufficient speaking practice, limited access to mentors, and the absence of AI systems capable of assessing conceptual explanations rather than only language fluency.
3. TRIGGERS TO ACT Users seek the platform when preparing for exams, viva voce, technical interviews, classroom presentations, or public speaking sessions. The need for instant AI-based evaluation, confidence building, and improvement in conceptual understanding motivates them to use VBCUA.	7. BEHAVIOR Users frequently watch educational videos, attend online courses, practice explaining concepts aloud, participate in mock interviews, and search for AI-powered learning tools. They repeatedly practice until they become confident in their explanations. Gemini AI, and offers an interactive dashboard for monitoring financial recovery. This enables borrowers to manage debt more effectively, negotiate confidently	10. YOUR SOLUTION The Voice-Based Concept Understanding Analyser (VBCUA) provides an AI-powered solution that allows users to upload or record spoken explanations of concepts. The system uses OpenAI Whisper for speech-to-text transcription, Sentence-BERT for semantic similarity analysis, and Librosa for audio feature extraction. It evaluates conceptual understanding, communication quality, and speech

	with lenders, and make informed financial decisions through a secure and user-friendly platform.	characteristics, generates an overall understanding score with qualitative feedback, and provides downloadable PDF reports to help users continuously improve their learning and presentation skills. making repayment decisions.
4.EMOTIONS Before using the platform, users often feel nervous, uncertain, and lack confidence about whether they truly understand a concept. After receiving AI-generated evaluation and feedback, they feel more confident, motivated, and better prepared for academic and professional assessments.		